Which is more accurate, LIGHT v0, v0_1 or v0_2

The three versions on the same tests. v0 is version 0.1.0, v0_1 is version 0.1.1 and v0_2 is version 0.1.2, 29 September 2026.

Bottom line. v0_2 is the best of the three models. It is more accurate than v0, it scores level with v0_1, and it is the only version whose default network rests on measurements and theory alone.

v0 falls behind because its default network holds n_e fixed and has no charge chemistry, and under Tahoma's beam it gives the gain at $\times 0.32$ the measured value. v0_1 and v0_2 give the same numbers on every kinetic dataset, every pure-gas check and the electron closure. At Tahoma they trade places. v0_2 gives the formation efficiency at $\times 0.99$, the nearest of the three, and the gain at $\times 0.89$. v0_1 comes closer on the gain, $\times 0.93$ by default and $\times 0.95$ as ML scores it, but only by leaving measurements behind. Its default network has no Kr_2^* route and so leaves out Kr^* + 2Kr, a measured channel that takes Kr^* from the harpoon reaction. ML's configuration adds the route with Kannari's estimates for Kr_2^* + F and Kr_2^* + Kr_2^*, two rates that have never been measured, and v0_2 with those two estimates gives v0_1's 6.10 %/cm. Under the predictive-only policy an estimate may not set a rate. v0_2 also gives every unmeasured or disputed rate as a band, so it says how far its answer can move, up to a factor 2.7 through Kr_2^* + F and down by 66 % through the trimer's temperature law across the notes' ranges.

Most of the gain v0_2 still misses at Tahoma lies inside its Kr_2^* + F band, whose top gives $\times 0.97$. A measurement of that rate would show whether the gap is that reaction.

v0_1 kept v0's KrF(B,C) block and every formation link, and v0_2 kept v0_1's, so wherever they run the same chemistry they score alike, and Table 1 lists every test. On the 300 K kinetic datasets the three agree to the digit. Under Tahoma's electron beam, with the ion and F_2-electron chemistry on, n_e free and the Kr_2^* group with Kannari's estimates in it, v0 gives the small-signal gain at $\times 0.96$ the measured 6.39 %/cm and v0_1 at $\times 0.95$. Yu and Setser's C/B ratio between 230 and 300 K improves from -13.6 to -8.4 %, but v0_1 and v0_2 book their B-C gap, which rests partly on that ratio, so the improvement is not an independent test.

The difference between v0 and the others is what each runs by default. v0's basic network holds n_e at $10^{15}\,\mathrm{cm}^{-3}$ and has no charge chemistry, so every ion it makes stays an ion. Under Tahoma's beam, which makes ion pairs, that network gives the gain at $\times 0.32$ the measured value, and $\times 0.32$ with the Kr_2^* group added. v0's overview scored $\times 0.96$ because ML ran v0 with its optional ion, F_2-electron and Kr_2^* groups on and n_e free, and the overview did not say so. v0_1 made that chemistry its default, and its default network gives $\times 0.93$. v0_2's gives $\times 0.89$.

Between v0_1 and v0_2 the difference at Tahoma is the Kr_2^* route. v0_2 carries its measured rows, Ueda's Kr^* + 2Kr, the 270-ns lifetime and Quigley and Hughes' Kr_2^* + F_2. Kr_2^* + F $\rightarrow$ KrF(B) + Kr and Kr_2^* + Kr_2^* Penning ionization have no measurement, so v0_2 leaves them at zero and gives the gain at $\times 0.89$. v0_1 as ML scores it, with Kannari's estimates for those two, gives $\times 0.95$, and v0_2 with the same two estimates gives $\times 0.95$, so those two rates are the whole difference. On the formation efficiency the order turns, $\times 0.99$ for v0_2 against $\times 1.06$ for v0_1, and the saturation intensities agree, 1.54 and 1.55 $\mathrm{MW/cm^2}$. Over the posterior of the B/C constants v0_2's gain spans 4.99 to 6.18 %/cm and v0_1's 5.38 to 6.63. Tahoma's gain would favour a nonzero Kr_2^* + F and its formation efficiency the zero, but Tahoma is device data, so it scores that rate and never sets it. At the top of its band v0_2 gives the gain at $\times 0.97$.

Two unmeasured quantities dominate what all three versions leave open. One is the trimer's temperature law above 425 K. At Tahoma's 446 K the weak $T_g^{-1/2}$ law gives v0_2's gain at $\times 0.44$ and v0_1's at $\times 0.47$, so Tahoma's gain counts against that law, as a score, and at the reference point it takes v0_2's 248-nm share from 4.67 to 2.22 %. The other is Kr_2^* + F, whose band takes the same share from 4.67 to 9.50 %. Each of v0_2's other bands moves it by 0.21 points or less. v0_2's changes are in `LIGHT/v0_2/README.md`, and v0_1's accuracy is written up in `docs/accuracy_overview_v0_1.pdf`.

test	measured	v0	v0_1	v0_2
KrF(B) decay against Kr and F_2, Eden 1978		rms 2.3 and 3.2 %	rms 2.3 and 3.2 %	rms 2.3 and 3.2 %
KrF(B,C) slow decay, Gadomski 1992		rms 6.1 and 4.2 %	rms 6.1 and 4.2 %	rms 6.1 and 4.2 %
C/B emission ratio at 300 K, Yu and Setser 1990	0.16 ± 0.02	-0.5 %	-0.5 %	-0.5 %
KrF(B,C) decays, Xu 1993 Figs. 5 to 7		rms 11.7 %	rms 11.7 %	rms 11.7 %
decay constants Xu 1993 printed	5.0×10^{7} and 6.2×10^{7} s^{-1}	$\times0.91$ and $\times0.97$	$\times0.91$ and $\times0.97$	$\times0.91$ and $\times0.97$
C-A decays, Hakuta 1990		rms 19.3 and 34.1 %	rms 19.3 and 34.1 %	rms 19.3 and 34.1 %
τ_B, Burnham and Searles 1977	9.0 ± 0.6 ns	-22.3 %	-22.3 %	-22.3 %
C/B ratio 230 K over 300 K, Yu and Setser	1.30 ± 0.06	-13.6 %	-8.4 %, not independent	-8.4 %, not independent
Kr^* + Kr^+ made per eV, pure Kr and Ar, Saito 2003		$\times1.02$ and $\times1.03$	$\times1.02$ and $\times1.03$	$\times1.02$ and $\times1.03$
$Kr_2^+ + e^-$ recombination, Shiu and Biondi 1977		$\times0.87$ to $\times1.08$	$\times1.00$, the input	$\times1.00$, the input
$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$, Smith et al. 1972	2.74×10^{-31} to 3.3×10^{-31}	2.5×10^{-31}	2.4×10^{-31}, a mean with two others	2.4×10^{-31}, the same mean
VUV yield of pure Kr at low pump, Morozov 2008	0.42 ± 0.05	0.57	0.57	0.57
electron closure at Tahoma, T_e and n_e against the kinetic loop		within 0.7 and 2.3 %	within 0.7 and 2.5 %	within 0.7 and 2.5 %
Tahoma small-signal gain g_0, each version's default network	6.39 %/cm	2.05, $\times0.32$	5.94, $\times0.93$	5.69, $\times0.89$
the same with the Kr_2^* group		2.04, $\times0.32$	6.10, $\times0.95$	in the default
as ML scores each version, with ion and F_2-electron chemistry, Kr_2^* and free n_e		6.11, $\times0.96$	6.10, $\times0.95$	5.69, $\times0.89$, the default
the same without the estimated $Kr_2^* + F \rightarrow KrF$		5.71, $\times0.89$	5.69, $\times0.89$	as above
with Kannari's $Kr_2^* + F$ and $Kr_2^* + Kr_2^*$, and at the top of the $Kr_2^* + F$ band				6.10, $\times0.95$, and 6.23, $\times0.97$
over the posterior of the B/C constants		5.83 [5.42, 6.63]	5.79 [5.38, 6.63]	5.39 [4.99, 6.18]
on each version's own power input, no beam		6.60, $\times1.03$	6.56, $\times1.03$	5.53, $\times0.87$
Tahoma saturation intensity I_{sat}	1.4 ± 0.1 MW/cm^2	1.54, $\times1.10$	1.54, $\times1.10$	1.55, $\times1.11$
Tahoma formation efficiency $g_0 I_{sat}/P_{in}$	0.27	0.284, $\times1.05$	0.285, $\times1.06$	0.268, $\times0.99$

Table 1: v0, v0_1 and v0_2 on the same tests. A percentage is the model's deviation from the data, signed for a single value and rms for a set. A $\times$ factor is the model over the measurement. Tahoma's gain is in %/cm. For v0_1, ML's configuration is its default network with the Kr_2^* group, and for v0_2 it is its default network. The two recombination rows compare v0_1 and v0_2 with the measurements they were built from.

Where the versions differ without data

At the notes' reference point, 1 atm of Kr with 0.5 % F_2 pumped at 300 kW cm^{-3} for 1 μs, no measurement exists, and there the defaults differ too. v0 sends 5.24 % of the deposited energy out at 248 nm, v0_1 4.73 % and v0_2 4.67 %. Table 2 takes the gaps apart. Between v0 and v0_1 the default network accounts for 0.39 of the 0.51 points. v0 holds n_e at 10^{15} cm^{-3}, where v0_1's free n_e runs at 1.95×10^{14} cm^{-3} while F_2 lasts, so electrons quench 19.0 % of the KrF in v0 and 6.8 % in v0_1. v0 also leaves 27.1 % of the deposited energy in ions that never recombine. v0_1 returns that energy through recombination, mostly as heat, and its gas ends at 762 K where v0's ends at 604 K. The values v0_1 changed account for the other 0.12 points. Only the B-A branch moves the 248-nm share by more than a few hundredths, since it sends 3.5 % of B's emission to 272 nm.

v0_2 differs from v0_1 by 0.07 points at 248 nm, but its gas ends 113 K cooler. After F_2 burns out, the Kr* that v0_1 turns into Penning ionization and heat forms Kr_2^* in v0_2 and leaves as 147-nm light, 20.0 % of the deposited energy. With Kannari's two estimates v0_2 gives 9.17 %, what v0_1 gives with its optional Kr_2^* group (9.17 % in v0_1's overview).

model at the reference point	248 nm, % of E_{dep}	147 nm, % of E_{dep}	T_g at the end, K
v0 by default, n_e held at 10^{15} cm^{-3}, no charge chemistry	5.24	0.0	604
v0 with its ion and F_2-electron groups, n_e free	4.85	0.0	762
v0_1	**4.73**	**0.0**	**762**
v0_1 without the B-A branch	4.90	0.0	762
v0_1 without F + F + Kr	4.69	0.0	762
v0_1 with v0's Kr^+ conversion and Kr_2^+ recombination	4.75	0.0	762
v0_1 without the ion-ion temperature law	4.73	0.0	763
v0_1 with REBUILD's KrF and Kr_2F^* energies	4.73	0.0	763
v0_2	**4.67**	**20.0**	**649**
v0_2 with Kannari's Kr_2^* + Kr_2^*	4.65	10.8	704
v0_2 with Kannari's Kr_2^* + F and Kr_2^* + Kr_2^*	9.17	1.3	724

Table 2: The reference point, one change at a time. The rows below v0_1 each undo one of its changes. The rows below v0_2 give its two unmeasured Kr_2^* rates Kannari's estimates, and the last is v0_1 with its optional Kr_2^* group.

v0_2 carries every rate that is known only as a published spread, or not at all, as a band. A measured rate keeps its measurement as the default and spans the other published values. A rate with no measurement is zero by default and runs to the largest published value, which is no bound. Table 3 runs each band alone at its two ends. Two of the 8 matter for the laser line, Kr_2^* + F and the trimer law. The Kr_2F^* quench bands raise the 248-nm share through the gas temperature, since their energy goes to heat instead of 420-nm light and the hotter gas slows the trimer.

Table 4 runs the three defaults over the notes' ranges, where no measurement exists either. v0 and v0_1 differ by up to 91 % of v0_1's value, and v0_1 and v0_2 by up to 11 %. Kr_2^* + F at the top of its band raises v0_2's share by up to a factor 2.7, and the trimer's weak law lowers it by up to 66 %. Those two bands reach farther than the spread between the versions at 10 of the 15 points. The spread is the wider only where v0_2's gas ends at 1173 K or more, beyond every rate law's data. No version can be preferred there on its numbers.

rate	v0_2's value	band	248 nm	147 nm	T_g end, K
$Kr_2^* + F \rightarrow KrF(B) + Kr$	0	0 to 1.0×10^{-9} $cm^3\,s^{-1}$	4.67 to 9.50	20.0 to 0.5	649 to 726
$Kr_2^* + Kr_2^* \rightarrow Kr_2^+ + 2Kr + e^-$	0	0 to 5.0×10^{-10} $cm^3\,s^{-1}$	4.67 to 4.65	20.0 to 10.8	649 to 704
$Kr_2F^* + F \rightarrow 2Kr + 2F$	0	0 to 1.8×10^{-10} $cm^3\,s^{-1}$	4.67 to 4.87	20.0 to 19.9	649 to 670
$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$, 300 K	1.7×10^{-32}	1.7×10^{-32} to 5.5×10^{-32} $cm^6\,s^{-1}$	4.67 to 4.52	20.0 to 28.0	649 to 601
$Kr_2^* + F_2$, both channels	2.1×10^{-10}	2.1×10^{-10} to 5.0×10^{-10} $cm^3\,s^{-1}$	4.67 to 4.60	20.0 to 19.1	649 to 654
$Kr^* + Kr^* \rightarrow Kr^+ + Kr + e^-$	5.0×10^{-10}	1.5×10^{-10} to 5.0×10^{-10} $cm^3\,s^{-1}$	4.87 to 4.67	25.0 to 20.0	616 to 649
$Kr_2F^* + F_2$	1.2×10^{-10}	1.2×10^{-10} to 4.3×10^{-10} $cm^3\,s^{-1}$	4.67 to 4.88	20.0 to 19.9	649 to 663
the trimer's $(T_g/300)^n$	-3.32	-3.32 to -0.50	4.67 to 2.22	20.0 to 19.5	649 to 658

Table 3: v0_2's bands at the reference point. Each row runs one band at its low end and then at its high end, the rest at their defaults, and gives the percentages of E_{dep} and the final gas temperature at the two ends. The unmeasured rates start at zero.

p	P	248 nm, % of E_{dep}					T_g at the end
atm	kW cm^{-3}	v0	v0_1	v0_2	v0_2, Kr_2^* + F at 10^{-9}	v0_2, trimer $T_g^{-1/2}$	v0_2, K
0.1	100	3.73	3.46	3.47	3.55	3.39	1173
0.1	300	1.25	0.80	0.80	0.81	0.79	2434
0.1	500	0.75	0.39	0.39	0.40	0.39	3458
0.25	100	8.43	9.22	9.23	10.33	8.07	826
0.25	300	2.83	2.39	2.40	2.61	2.15	2044
0.25	500	1.71	1.21	1.21	1.30	1.10	3102
0.5	100	11.31	14.21	13.84	16.13	9.25	532
0.5	300	4.38	4.11	4.13	5.86	2.93	1191
0.5	500	2.63	2.24	2.26	3.03	1.65	1928
1	100	4.79	5.73	5.42	5.84	3.28	440
1	300	5.24	4.73	4.67	9.50	2.22	649
1	500	3.09	2.72	2.71	7.18	1.33	915
2	100	1.00	1.25	1.17	1.28	0.83	378
2	300	2.30	2.45	2.17	2.64	0.89	519
2	500	2.81	2.31	2.12	4.48	0.73	604

Table 4: The three defaults over the notes' pressure and power ranges, 0.5 % F_2 for 1 μs. The sixth and seventh columns put v0_2's two wide bands at the ends that move the share most.

Generated from LIGHT v0 (0.1.0), v0_1 (0.1.1), v0_2 (0.1.2) and ML by `docs/make_overview.py`, `docs/make_overview_v0_1.py`, the two map scripts and `docs/REVIEW/make_review.py`. No number in this document is typed by hand.